



Security Audit & Assessment MECR2123

ASSIGNMENT (10%)

(Due date: 20 May 2026)

Session 2025/2026-2

Lecturer: Assoc. Prof. Ts. Dr. Siti Hajar Othman

Instruction: Answer all questions

Submission: Submit through your e-Learning: “*Assignment 1*”.

Student Name:	KEERTENNAH DEVI A/P PONNAMBALAM
Student Metric No:	MEC254026
<i>I admit I made this assignment</i>	Sign: 

INSTRUCTION:

Run the following task. Then, produce a report based on the step-by-step process you conducted.

Download the Vosviewer Software

Please download Free the Vosviewer software, get familiar with the software

Visualize research trend based on domain keywords selected

Based on the **TOPIC LIST (below)**:

- Select TWO from the following topics,
- Determine relevant keywords based on the topic,
- Enter PSZ Database Online, go to Scopus database;
- Populate journals from Scopus website;
- Start visualize research trends based on your topic and keywords.
- Snapshot the 'Network Visualization' and 'Density Visualization'.
- Identify concepts clustered by Vosviewer.

TOPIC LIST:

- 1) Data Center Security Audit
- 2) Switch Security Audit
- 3) Firewall Security Audit
- 4) Cloud computing Security Audit
- 5) Operating System Security Audit
- 6) Mobile Security Audit
- 7) Web Server Security Audit
- 8) Web Application Security Audit

E.g.: 3) Firewall Security Audit and 6) Mobile Security Audit

E.g.:

Keyword: Firewall Security Audit

How many journal papers:

The screenshot shows a search interface with two input fields. The first field, labeled 'Search within', has a dropdown menu showing 'Article title, Abstract, Keywords'. The second field, labeled 'Search documents *', contains the text 'web AND application AND security AND audit'. Below the search fields are buttons for '+ Add search field', 'Reset', and 'Search Q'. A navigation bar includes links for 'Documents', 'Preprints', 'Patents', 'Secondary documents', and 'Research data 71'. A 'Beta' badge is visible. The results section shows '203 documents found' and an 'Analyze results 71' link. A toolbar includes options for 'All', 'Export', 'Download', 'Citation overview', 'More', 'Show all abstracts', and 'Sort by Date (newest)'. The results are displayed in a table with columns for 'Document title', 'Authors', 'Source', 'Year', and 'Citations'. Two results are shown, both marked as 'Conference Paper'.

	Document title	Authors	Source	Year	Citations
☐ 1	Conference Paper Design and implementation of security policy for HEPS container computing platform	Hu, Q., Cui, T., Xu, J., Yan, T.	Proceedings of Science, 434, 002	2023	0
Show abstract Related documents					
☐ 2	Conference Paper Real-time Tracking of Auditing Process Progress with a Customizable Application for Cybersecurity Standards	Zelmati, M., Oulqaid, Z., Elouadi, A.	Proceedings - 10th International	2023	0

Snapshot the Cluster you get from Vosviewer:

E.g.:

Cluster 1 (42 items)
access
access control
addition
architecture
article
attack
audit
audit trail
authentication
business
client
code
cross site
database
developer
end
event
example
goal
integrity
internet
issue

Step 1: Choose Two Topics

Topics Selected:

- 1) Data Center Security Audit
- 5) Operating System Security Audit

Step 2: Define Keywords for Scopus Search

- 1) Data Center Security Audit

Keywords used: data center AND security AND audit

- 5) Operating System Security Audit

Keywords used: operating system AND security AND audit

Step 3: Run the Search on Scopus via PSZ Database

- 1) Data Center Security Audit

Total journal papers found: 183

Language: English

Advanced query (●)

Search within
Article title, Abstract, Keywords

Search documents *
data center AND Security AND Audit

+ Add search field Reset Search Q

Documents **Beta** Preprints Secondary documents

183 documents found [Analyze results ↗](#)

☐ All Export Download Citation overview ... More Show all abstracts Sort by Date (newest) [Grid](#) [List](#)

	Document title	Authors	Source	Year	Citations
☐ 1	Article • Open access Implementation and sustainability of an innovative ICU e-diary	Rose L., Welch A., Okelana K.,... Susser K., Meyer J.	Intensive and Critical Care Nursing , 93, 104266	2026	3
Show abstract View at Publisher ↗ Related documents					
☐ 2	Article Reimagining third places: the role of GenAI robots in shaping interaction and trust in a polarized society	Paluch S., Wirtz J., Pitardi V., Kunz W.H.	Journal of Services Marketing , 40(1)	2026	1
Show abstract View at Publisher ↗ Related documents					
☐ 3	Article • Open access An Anti-Sheriff Cybersecurity Audit Model: From Compliance Checklists to Intelligence-Supported Cyber Risk Auditing	Rananga N., Venter H.S.	Applied Sciences Switzerland , 16(5), 2315	2026	0
Show abstract View at Publisher ↗ Related documents					

Figure 1

5) Operating System Security Audit

Total journal papers found: 313

Language: English

Search within: Article title, Abstract, Keywords

Search documents: operating system AND security AND audit

+ Add search field Reset Search

Documents Preprints Secondary documents

313 documents found Analyze results

All Export Download Citation overview More Show all abstracts Sort by Date (newest)

	Document title	Authors	Source	Year	Citations
1	A comprehensive bluetooth security audit framework for IoT devices	Kavisankar L., Aslam N., Vemuri A., ... Kumar Shah N., Venkatesan S.	Computers and Security, 164, 104840	2026	0
Show abstract View at Publisher Related documents					
2	Operationalizing Resilience in Critical Logistics Infrastructures: A Reliability-Based Decision Support System Grounded in Eurocode Standards	Moyano Retamero J., Camarero Orive A.	Systems, 14(2), 191	2026	0
Show abstract View at Publisher Related documents					
3	A Sensitive Information Masking-Based Data Security Auditing Method for Chinese Linux Operating System	Ma W., Guo H., Xia A., Mao X.	Electronics Switzerland, 15(1), 86	2026	0
Show abstract View at Publisher Related documents					

Figure 2

Step 4: Export Data for Vosviewer

- Select all results
- Click Export to CSV or RIS

Step 5: Import into Vosviewer

1. Open Vosviewer and click create a new map
2. Choose "Create map based on bibliographic data"
3. Select "Read data from bibliographic database files" (Scopus)
4. Upload your exported file
5. Choose analysis type: "Co-occurrence" and "All keywords"
6. Set threshold and number of keywords
7. Click Finish to generate the visualization

Step 5: Capture Snapshots

1) Data Center Security Audit

- Network Visualization

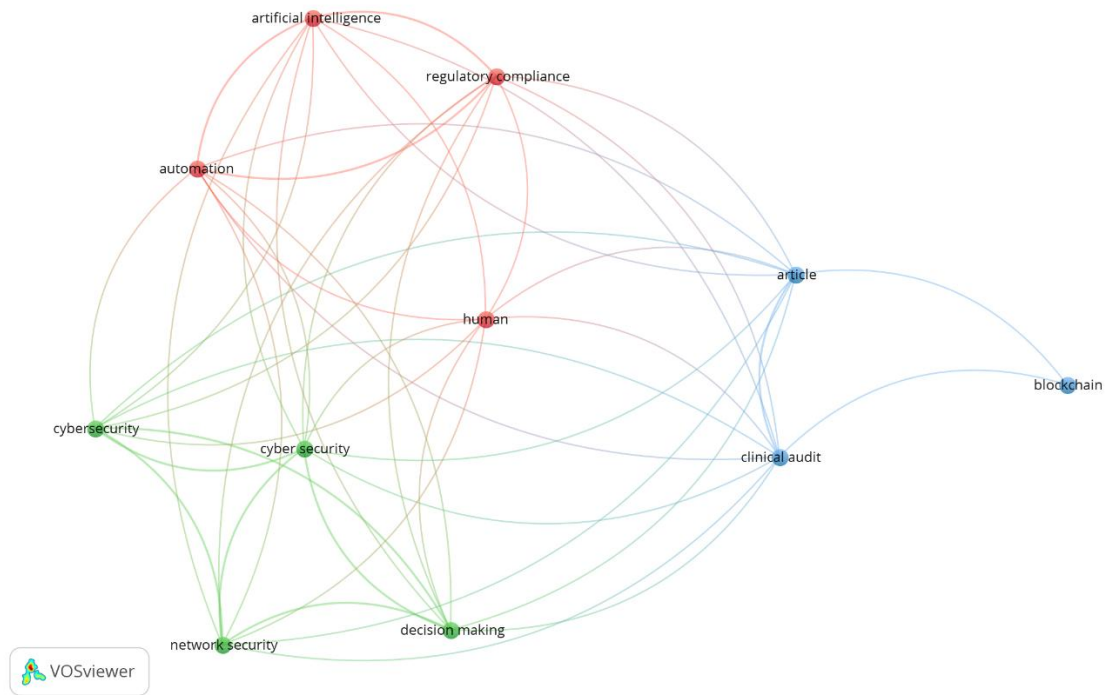


Figure 3

Clusters identified: Cluster 1 (Red), Cluster 2 (Green) and Cluster 3 (Blue)

Cluster 1 (4 items)
artificial intelligence
automation
human
regulatory compliance
Cluster 2 (4 items)
cyber security
cybersecurity
decision making
network security
Cluster 3 (3 items)
article
blockchain
clinical audit

Figure 4

- Density Visualization

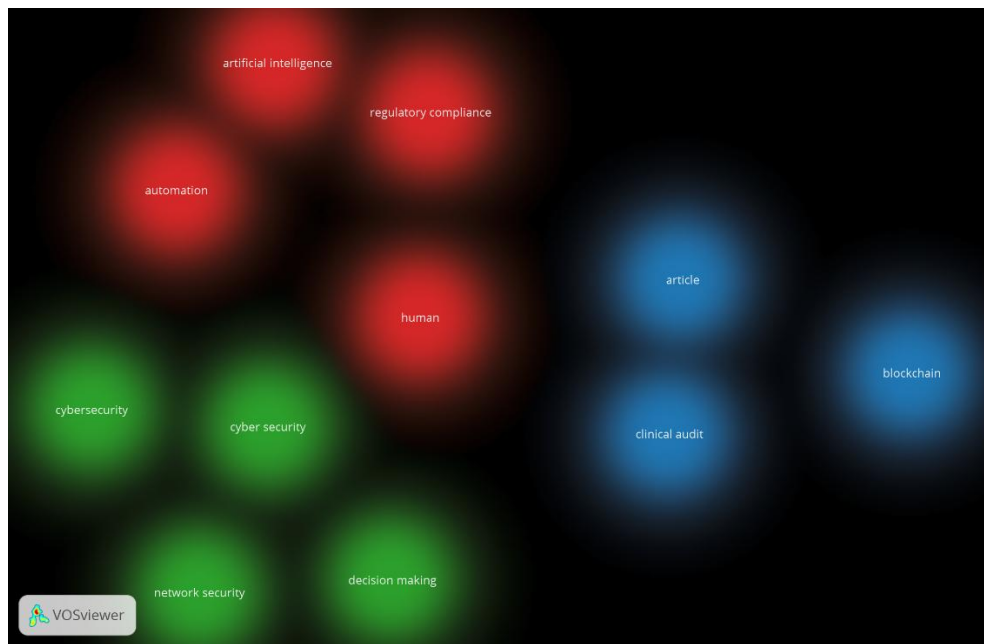


Figure 5

Observation:

Highest density is around cybersecurity and network security and these are the most established research themes. Medium density around artificial intelligence and automation, indicating growing interest in AI-driven audits. Lower density includes blockchain and clinical audit emerging topics. The presence of human and articles likely represents general bibliometric terms rather than core security concepts.

5) Operating System Security Audit

- Network Visualization

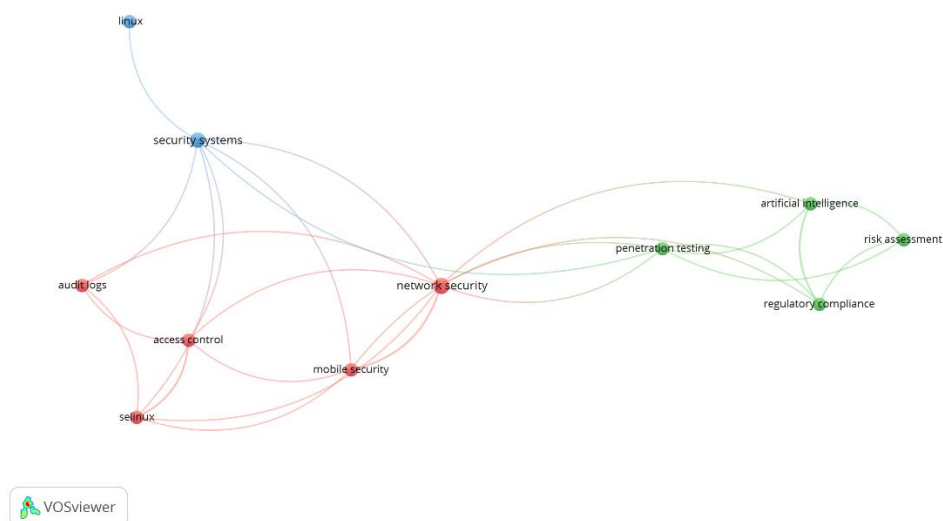


Figure 6

Clusters identified: Cluster 1 (Red), Cluster 2 (Green) and Cluster 3 (Blue)

11 items (3 clusters):
Cluster 1 (5 items)
access control
audit logs
mobile security
network security
selinux
Cluster 2 (4 items)
artificial intelligence
penetration testing
regulatory compliance
risk assessment
Cluster 3 (2 items)
linux
security systems

Figure 7

- Density Visualization

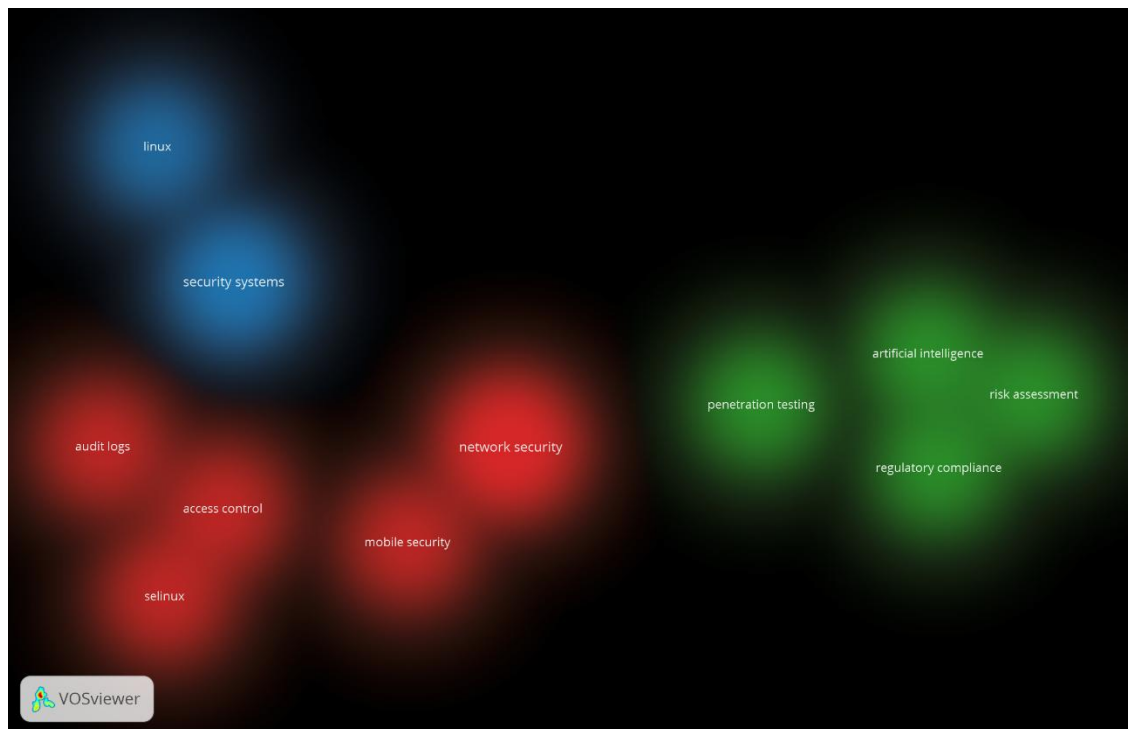


Figure 8

Observation:

The highest density is centered around linux, access control, and audit logs and these are the most mature and heavily researched topics in operating system security auditing. Medium density appears around network security and selinux (a Linux security module). Lower density includes mobile security and security systems as broader or slightly less focused terms. The density map confirms that Linux-based OS auditing is the dominant theme, while mobile OS security is present but less dense, suggesting it is a growing.

Conclusion

In this assignment, the Scopus database was successfully used with Vosviewer to visualize research trends on both topics which are Data Center Security Audit (183 papers) and Operating System Security Audit (313 papers).

The Data Center Security Audit had two clear primary themes around cybersecurity and network security and expressed interest in the areas of AI, automation, and blockchain. While "clinical audit" was present in this area, indicating some degree of cross-domain terminology transfer, the fundamental research in this domain is still related to technical security controls.

The most obvious cluster for the Operating System Security Audit is based on security methods and measures related to Linux Operating Systems (SELinux, Access Control and Audit Logs). A secondary cluster comprises Network Security, Mobile Security and Penetration Testing, while Risk Assessment and Regulatory Compliance appear to be supporting themes. The use of Artificial Intelligence in OS auditing is also present but is still in the early stages of development.

Overall, Vosviewer has successfully represented the core, mature and emerging research groups, allowing for a clear visual understanding of each domain's research landscape.